

Q1: According to Figure 1, a force P equal to 5 kilonewtons is applied to point C of the structure. Using the principle of conservation of energy, determine the vertical displacement of point C . ($AE = 64 \times 10^6 \text{ N}$)

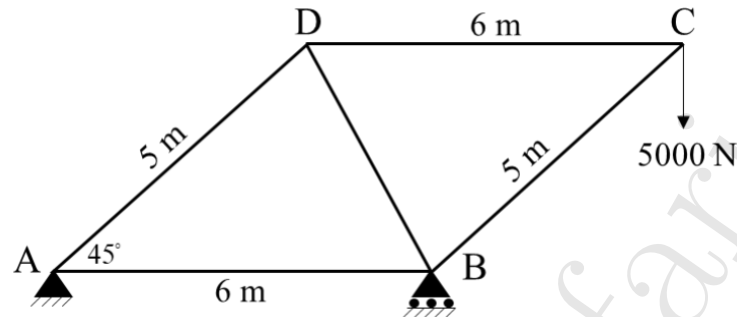


Figure 1: Q1's Truss

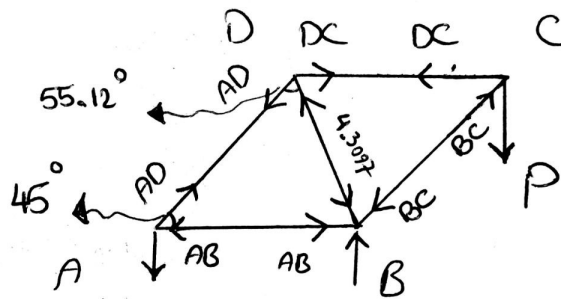


Figure 2: Truss's Forces

Static Equilibrium:

$$R_B = 1.5893P \uparrow (N), A_y = 0.5893P \downarrow (N)$$

Sine and Cosine rules:

$$DB = 4.3097 \text{ m}, \hat{ABD} = 55.12^\circ$$

Joint C:

$$BC = \sqrt{2}P \nearrow (N), DC = P \leftarrow (N)$$

Joint A:

$$AD = 0.8329P \nearrow (N), AB = 0.5893P \leftarrow (N)$$

Joint D:

$$BD = 0.7179P \nwarrow (N)$$

Strain Energy and External Work:

Truss Strain Energy

From previous calculations, for each element we have:

Element	F_i	L_i	$F_i^2 L_i$	Load
DC	$+P$	6	$6P^2$	Tensile
CB	$-\sqrt{2}P$	5	$6P^2$	Compressive
AB	$-0.5893P$	6	$2.0836P^2$	Compressive
DB	$-0.7197P$	4.3097	$2.2323P^2$	Compressive
AD	$+0.8329P$	5	$3.4687P^2$	Tensile

Table 1: Elements Loading

Displacement

$$\delta_c = 1.858 \text{ mm}$$

Q2: Consider the structure shown in Figure 3. Using the method of virtual work and neglecting the effects of shear deformation, determine the following:

- (a) Calculate the vertical displacement of point C.
- (b) Calculate the slope at point A.

The modulus of elasticity is $E = 1.9 \times 10^6 \left(\frac{\text{lb}}{\text{ft}^2} \right)$.

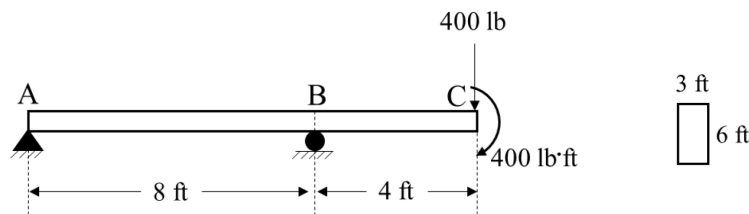


Figure 3: Q2's Beam

a. Vertical displacement:

Real Forces

Static Equilibrium:

$$R_A = 250 \text{ lb} \downarrow (\text{lb}), R_B = 650 \text{ lb} \uparrow (\text{lb})$$

A to B:

$$M(x_1) = -250x_1, 0 \leq x_1 \leq 8$$

C to B:

$$M(x_2) = 400x_2 + 400, 0 \leq x_2 \leq 4$$

Moment Distribution:

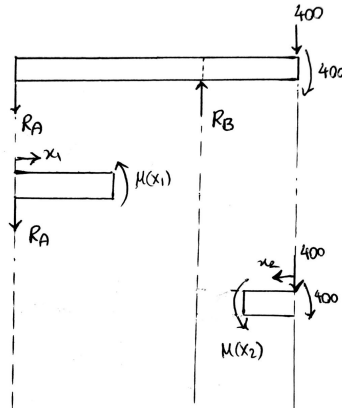


Figure 4: Beam Real Force Diagram

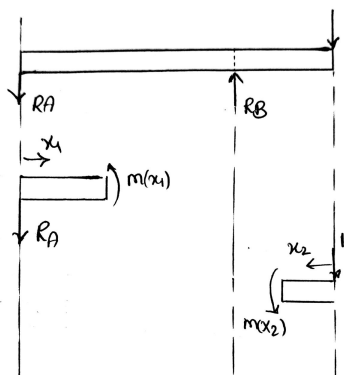


Figure 5: Beam Virtual Loading For Vertical Displacement Diagram

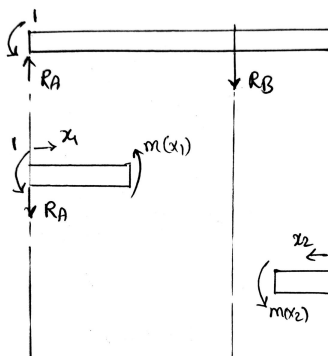


Figure 6: Beam Virtual Loading For Slope Diagram

Virtual Forces: Put downward unit load at C.

Static Equilibrium:

$$r_A = 0.5 \text{ lb} \downarrow (\text{lb}), r_B = 1.5 \text{ lb} \uparrow (\text{lb})$$

A to B:

$$\bar{m}(x) = -0.5x_1, \quad 0 \leq x_1 \leq 8$$

C to B:

$$\bar{m}(x_2) = x_2, \quad 0 \leq x_2 \leq 4$$

Moment Distribution:

Vertical displacement:

$$\Delta = 3.2229 \times 10^{-4} \text{ ft} = 3.8674 \times 10^{-3} \text{ inch}$$

b. slope at point A

Virtual moment: Put CCW unit moment at A.

Static Equilibrium:

$$r_A = \frac{1}{8} \text{ lb } \uparrow (\text{lb}), \quad r_B = \frac{1}{8} \text{ lb } \downarrow (\text{lb})$$

A to B:

$$\bar{m}_1(x_1) = \frac{1}{8}x_1 - 1, \quad 0 \leq x_1 \leq 8$$

C to B:

$$\bar{m}(x_2) = 0, \quad 0 \leq x_2 \leq 4$$

Moment Distribution:

slope at point A:

$$\Delta = 0.001489^\circ$$

Q3. The structure shown in Figure 7 is considered a frame. Assuming all members have the same flexural rigidity EI , use Castigliano's theorem to determine the vertical deflection of point C (consider only bending deformation).

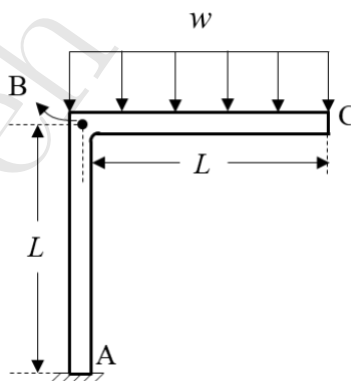


Figure 7: Q3's Frame

Castigliano's Equation:

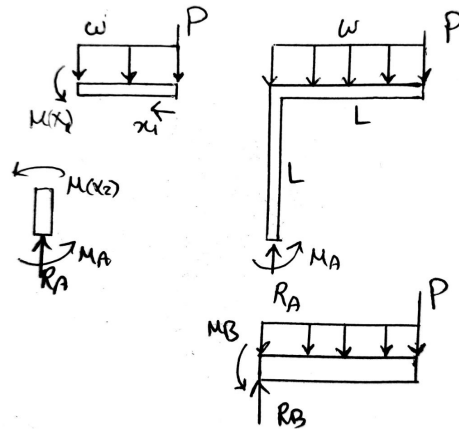


Figure 8: Frame Loading Diagrams

Static Equilibrium:

$$A_y = wL + P \uparrow, M_A = PL + \frac{1}{2}wL^2 \text{ ccw}$$

BC element Static Equilibrium:

$$B_y = wL + P \uparrow, B_A = PL + \frac{1}{2}wL^2 \text{ ccw}$$

BC element Load distribution: (right to left)

$$BC : M(x) = Px + \frac{1}{2}wx^2, 0_C \leq x \leq L_B$$

AB element Load distribution: (down to up)

$$AB : F(x) = wL + P, \quad M(x) = PL + \frac{1}{2}wL^2, \quad 0_A \leq x \leq L_B$$

Castigliano's Equation:

$$\Delta_C = \frac{5}{8EI}wL^4 + \frac{1}{AE}wL^2$$

Q4: Consider a cantilever beam subjected to a uniformly distributed load such that $p(x) = -p_0$, a point force F , and an applied moment C at the free end (see Figure 9). To solve this problem using the Rayleigh–Ritz method, the following trial function is proposed for estimating the beam deflection:

$$v(x) = a + bx + c_1x^2 + c_2x^3$$

- (a) Determine the unknown coefficients a, b, c_1, c_2 .
- (b) Compute the deflection at the free end of the beam.

$$E = 100 \text{ GPa}, \quad I = 10^{-7} \text{ m}^4, \quad L = 1 \text{ m}, \quad p_0 = 300 \text{ N/m}, \quad F = 500 \text{ N}, \quad C = 100 \text{ N} \cdot \text{m}.$$

a. Unknown Coefficients

Check BCs:

$$v(x) = c_1x^2 + c_2x^3, \quad v'(x) = 2c_1x + 3c_2x^2, \quad v''(x) = 2c_1 + 6c_2x$$

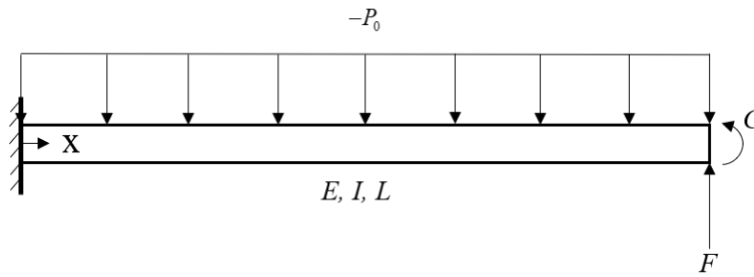


Figure 9: Q4's Beam

Rayleigh-Ritz method:

$$U = EI(2c_1^2L + 6c_1c_2L^2 + 6c_2^2L^3)$$

$$V = -F(c_1L^2 + c_2L^3) - C(2c_1L + 3c_2L^2) + P_0\left(\frac{1}{3}c_1L^3 + \frac{1}{4}c_2L^4\right)$$

by solve equation (II) with (I):

$$c_1 = \frac{1}{24EI}(12C + 12FL - 5L^2P_0), \quad c_2 = \frac{1}{12EI}(-2F + LP_0)$$

$$a = 0 \text{ (m)}, \quad b = 0 \text{ (1)}, \quad c_1 = 0.002375 \left(\frac{1}{m}\right), \quad c_2 = -\frac{1}{3}0.00175 \left(\frac{1}{m^2}\right)$$

b. Deflection at the free end:

$$v(L) = 0.00179 \text{ m} = 1.7917 \text{ mm}$$

Q5: The structure shown in Figure 5 is considered a frame. Compute all support reactions.

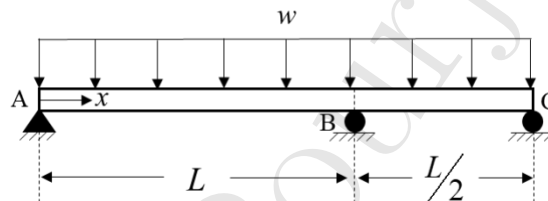


Figure 10: Q5's Beam

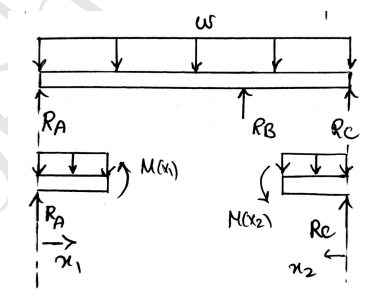
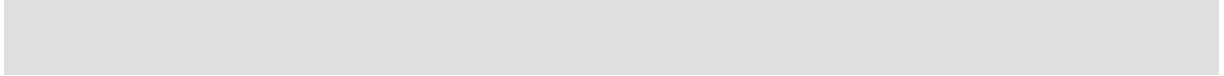


Figure 11: Beam Real Force Diagram

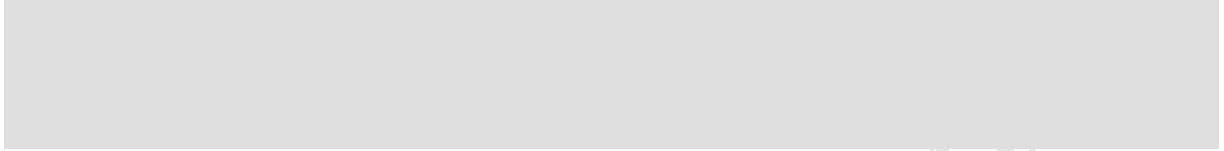
Static Equilibrium:

$$R_C = 2R_A - \frac{3}{4}wL, \quad R_B = \frac{9}{4}wL - 3R_A$$

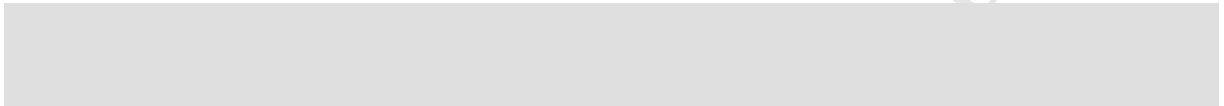
Section A to B:



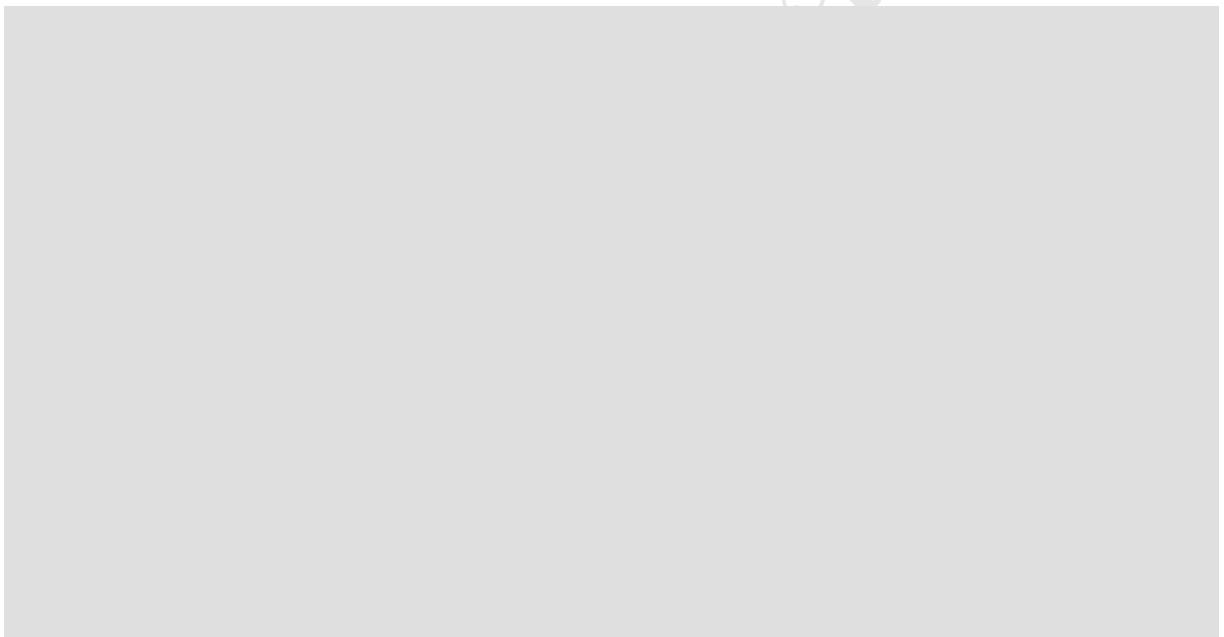
Section C to B:



Moment Distribution



Castigliano's Equation:



$$R_A = \frac{13}{32}wL, \quad R_B = \frac{33}{32}wL, \quad R_C = \frac{2}{32}wL$$